

Total no. of Pages 01

Roll no.

FIFTH SEMESTER B. Tech.
MID-SEMESTER EXAMINATION, SEPTEMBER 2023

Course Code- EAEPC 05

Course Title- Signals and Systems

Time- 1.5 Hours

Max. Marks- 25

Note: - Attempt all questions. Missing data/ information if any, maybe suitably assumed & mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Define a (i)Signal (ii)systems and classify them	2	CO 1
1b	Draw the function $\pi(2t+3)$ when $\pi(t) = 1$; for $t \leq 1/2$, 0 ; otherwise	3	
2a	Define following signals-(a) continuous time unit step and unit impulse (b) Energy and power signal?	2	CO1
2b	Determine whether the following signal is energy or power signal. And calculate its Energy or Power. $x(t) = e^{-2t}u(t)$.	3	
3a	What is meant by linear time invariant system?	2	CO2
3b	Draw the following signals. (i) $u(t) - u(t-10)$ (ii) $(1/2)^n u(n-1)$	3	CO1
4a	Define a causal system.	2	CO2
4b	Determine true or false in following conditions (i)if an LTI system is causal, it is stable. (ii) The inverse of a causal LTI system is always causal. (iii) If $h(t)$ is the impulse response of an LTI system and $h(t)$ is periodic and nonzero, the system is unstable.	3	CO2
5a	Explain memory and memoryless system with example.	2	CO2
5b	Consider the following pair of signals and determine whether there is a discrete time LTI system for which $y[n]$ is output when corresponding input is $x[n]$ (a) $x[n] = (1/2)^n$, $y[n] = (1/4)^n$ (b) $x[n] = (1/2)^n$, $y[n] = (1/2)^n u[-n]$	3	CO2